01.01 · THEORY + PAST-PAPER INDEX

Celestial Mechanics

From Newton's law of gravity to orbital motion

START HERE

This lecture builds orbital mechanics from the ground up. We begin with the force between two masses, then use conservation laws to understand circular orbits, ellipses, binaries, tidal disruption and the equilibrium points of a rotating system. Unless stated otherwise, bodies are treated as point masses and external perturbations are neglected.

LEARNING GOALS

✓	Translate Newton's law of gravitation into orbital acceleration and speed.
✓	Use a barycentre to describe two-body and binary-star motion.
✓	Connect Kepler's laws, orbital energy and angular momentum.
✓	Interpret the Roche limit and the five Lagrange points.

1 · Gravity is a central force

Newton's law of universal gravitation says that two point masses attract one another. The force grows with either mass and falls with the square of their separation. It points along the line joining the bodies, so it is a central force.

$$F = G\,m_1 m_2 / r^2 \cdot g(r) = GM / r^2$$

Here $G = 6.674 \times 10^{-11}$ m³ kg⁻¹ s⁻², r is the separation, F is the magnitude of the force, and g(r) is the gravitational acceleration produced by a central mass M. The force on the smaller body is directed inward. Newton's third law gives an equal and opposite force on the central body; both objects move.

FIELD AND POTENTIAL

Outside a spherical, non-rotating body, its gravity is the same as if all its mass were concentrated at its centre. The gravitational potential per unit mass is $\Phi = -GM/r$, chosen to be zero infinitely far away. Potential energy is $U = m\Phi$. The negative sign marks a bound configuration.

For an isolated system, the total linear momentum and angular momentum are conserved. These conservation laws are more useful than treating gravity as a memorized list of orbit formulas: they explain why a comet speeds up near its star and why the two stars in a binary share one period.

2 · The barycentre and the two-body problem

The centre of mass, or barycentre, is the mass-weighted average position. For two bodies of masses m_1 and m_2 separated by distance r, the barycentre lies on the line between them. If their distances from it are r_1 and r_2, then $m_1 r_1 = m_2 r_2$ and $r_1 + r_2 = r$.

$$r_1 = [m_2 / (m_1 + m_2)]\ r \cdot r_2 = [m_1 / (m_1 + m_2)]\ r$$

The more massive body is closer to the barycentre. A planet does not sit perfectly still while a satellite orbits it; the planet and satellite both orbit their common barycentre. For a binary star, measuring both orbital sizes and the period can reveal the mass ratio and total mass.

RELATIVE MOTION

Instead of following both position vectors separately, define the separation vector $r = r_2 - r_1$. The relative coordinate obeys the same inverse-square orbit equation as a test particle around a central mass $M = m_1 + m_2$. For a circular binary with separation a, the period is $P^2 = 4\pi^2 a^3/[G(m_1 + m_2)]$.

A useful physical check is that the two bodies have the same angular speed and period, but not the same linear speed. Since $v_1/v_2 = r_1/r_2 = m_2/m_1$, the lighter object moves on the larger, faster orbit.

3 · Circular orbits, orbital periods and escape

In a circular orbit, gravity supplies the centripetal acceleration. Equating GM/r^2 with v^2/r gives the orbital speed. The period follows by dividing the orbit's circumference by that speed.

$$v_{circ} = \sqrt{(GM/r)} \cdot P = 2\pi\sqrt{[r^3/(GM)]}$$

The formulas assume a negligible-mass satellite orbiting a central mass M. For two comparable masses, replace M by $m_1 + m_2$ and use their separation as the relative-orbit radius. The third-law form $P^2 \propto a^3$ means that distant orbits take much longer: doubling a increases the period by 2^(3/2), not by two.

$$E = K + U = -GMm/(2r) \cdot v_{esc} = \sqrt{(2GM/r)} = \sqrt{2}\ v_{circ}$$

A circular orbit is bound because its total mechanical energy is negative. Escape speed is the launch speed for which the final speed approaches zero at infinity, with no further propulsion or energy loss. It is larger than circular speed at the same radius by $\sqrt{2}$. An orbiting astronaut is not beyond gravity; the astronaut and spacecraft are in continuous free fall together.

DIMENSIONAL AND LIMITING CHECKS

GM/r has units of $m^2\ s^{-2}$, so its square root is a speed. Increasing M raises orbital speed; increasing r lowers it. Escape speed tends to zero far from the mass. These checks catch many algebra and unit errors.

4 · Ellipses and Kepler's laws

A gravitationally bound orbit is an ellipse with the attracting mass at one focus. The closest point is periapsis and the farthest is apoapsis; for a star, the names are periastron and apastron. The semimajor axis a is half the longest diameter, and eccentricity e measures how stretched the ellipse is, from $e = 0$ for a circle to $e < 1$ for a bound ellipse.

$$r_{peri} = a(1 - e) \cdot r_{apo} = a(1 + e)$$

Kepler's first law describes the ellipse. His second law says that the line from the focus to the orbiting body sweeps out equal areas in equal times. Gravity exerts no torque about the focus, so angular momentum is conserved; the body must move faster when it is closer in order to sweep the same area.

$$\frac{dA}{dt} = \text{constant} \cdot v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)$$

The second expression is the vis-viva equation. At periapsis and apoapsis the velocity is perpendicular to the radius, so it also follows that $v_{peri}/v_{apo} = r_{apo}/r_{peri}$. Kepler's third law relates period to semimajor axis: $P^2 = 4\pi^2 a^3/(GM)$ for a negligible-mass orbiting body, or with $M = m_1 + m_2$ for the relative orbit of two bodies.

ENERGY VIEWPOINT

For any bound two-body Kepler orbit, total orbital energy depends on a, not on the body's location along the ellipse: $E = -GMm/(2a)$. At periapsis kinetic energy is greatest and gravitational potential energy is lowest; at apoapsis the reverse is true.

5 · Tides, the Roche limit and Lagrange points

A planet's pull is stronger on the near side of a moon than on its far side. This difference in gravitational acceleration is a tidal force. A self-gravitating moon can remain intact only if its own gravity and material strength can resist the differential pull. A fluid satellite can be disrupted inside an approximate Roche distance.

$$d_{Roche} \approx 2.44 \, R_p \left(\rho_p / \rho_s\right)^{1/3}$$

Here R_p and ρ_p are the planet's radius and density, and ρ_s is the satellite's density. This coefficient applies to an ideal fluid satellite in a simplified circular orbit; rigidity, spin, internal structure and orbital eccentricity change the result. The key trend is robust: a larger density contrast pushes the disruption distance outward.

In a frame rotating with a two-body system, centrifugal effects combine with the two gravitational fields. The five Lagrange points are locations where the net effective force on a small test object is zero in that rotating frame. L1 lies between the two masses; L2 beyond the smaller mass; L3 on the far side of the larger mass; L4 and L5 lead and trail the smaller mass by 60 degrees.

EQUILIBRIUM IS NOT THE SAME AS STABILITY

L1, L2 and L3 are saddle points: a small displacement in some directions grows. L4 and L5 can be stable when $\mu = m_2/(m_1 + m_2)$ is below about 0.0385. The Sun–Jupiter and Earth–Moon systems satisfy this idealized condition. Real perturbations and radiation pressure can still alter a particle's motion.

6 · A reliable orbital calculation

First draw the geometry and decide which distance is meant: altitude above a surface, distance from a centre, or separation between two orbiting bodies. Convert all quantities to SI units. Then identify whether the orbit is circular, elliptical or a two-body relative orbit, and state the mass that belongs in Kepler's law.

Use conservation of energy for speed changes and escape, angular momentum for areal motion and periapsis-to-apoapsis comparisons, and the barycentre for binary sizes. Keep intermediate units: a period computed from SI values emerges in seconds. Round only at the end, and test the answer against a limiting case such as $e \to 0$ or $m_2 \ll m_1$.

MODEL LIMITS

Kepler's two-body model is an approximation. Additional planets, non-spherical mass distributions, atmospheric drag, radiation pressure and relativistic corrections can perturb real trajectories. In introductory IOAA-style calculations, the statement normally tells you when those effects may be ignored.

PAST IOAA PROBLEM REFERENCES

The following questions are from official past IOAA problem sets.

Use each year, round and question label to locate the original problem in its official paper.

01	IOAA 2007 · TH 1.6
02	IOAA 2007 · TH 1.7
03	IOAA 2009 · TH P3
04	IOAA 2009 · TH P6
05	IOAA 2009 · TH P12
06	IOAA 2009 · TH P14
07	IOAA 2010 · TH S6
08	IOAA 2011 · TH S1
09	IOAA 2011 · TH S3
10	IOAA 2014 · TH P1
11	IOAA 2014 · TH L3
12	IOAA 2015 · TH S1
13	IOAA 2015 · TH S5
14	IOAA 2015 · TH S13
15	IOAA 2016 · TH T9
16	IOAA 2019 · TH 9
17	IOAA 2021 · TH Q3
18	IOAA 2021 · TH Q9
19	IOAA 2021 · TH Q11
20	IOAA 2021 · TH Q12

21 IOAA 2022 · TH 8
22 IOAA 2022 · TH 12
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